Dynamic Programming

	Key(0)	(1)	(2)
freq	34	8	50

Optimal Binary Search trees

$$\text{opt cost}(i,j) = \sum_{k=i}^{j} \text{freq}(k) + \min_{r=i}^{j} \left(\text{opt cost}(i, r-1), \text{opt cost}(r+1, j)\right)$$

Calculate optimal cost cost[0, n-1] for final result.

	0	1	2
0	34	50	142
1		8	66
2			50

opt cost[0,2] 42 + Min [opt((0,-1) + opt(1,1)]

[opt(0,0) opt(2,1)]

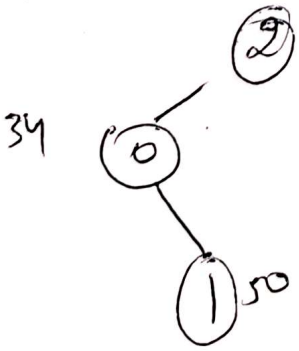
cost[1,2] = 58 + min (opt((1,0) + opt(2,2)]

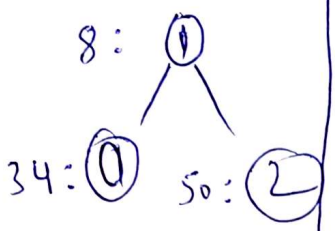
[opt(1,1) opt(3,2)]

opt[0,2] = 92 + min [opt(0,-1) + opt(1,2)] 66

+ [opt(0,0) + opt(2,2)] 84

[opt(0,1) + opt(3,2)] 50

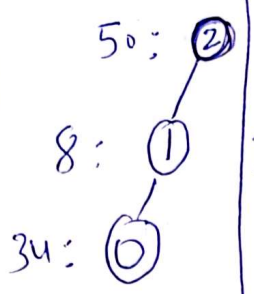




$8 \times 1 + 2(34 + 50)$

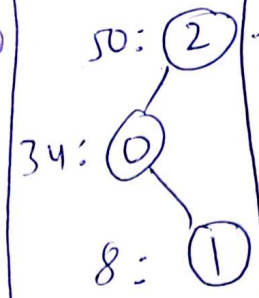
$\downarrow$

176



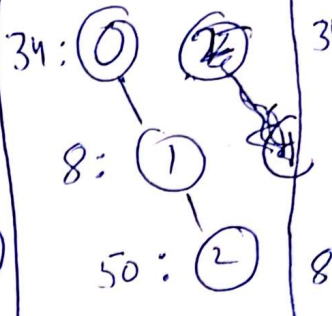
$50 + 16 + 34 \times 3$

168



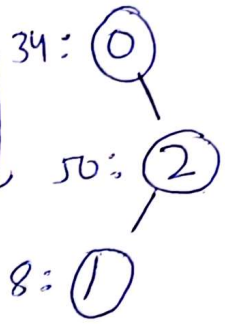
$50 + 68 + 8 \times 3$

142



$34 + 16 + 150 +$

200



$34 + 100 + 24$

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